

Instructor: Date:

Learner: Regional Branch:

Program: Center:

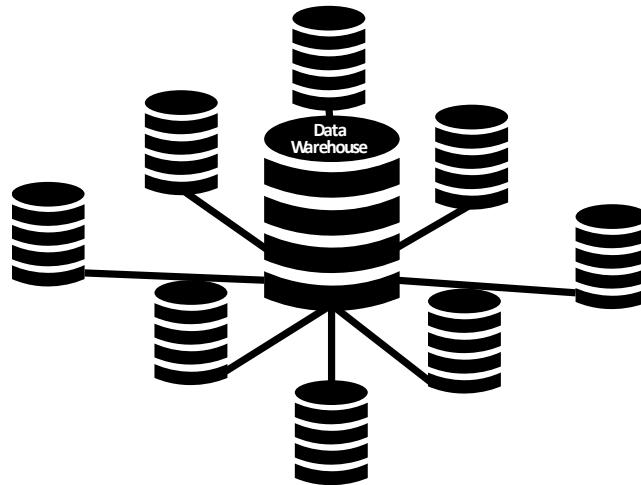
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Let's Start!

Historical Facts Behind the Concept of Data Warehousing



Learning Outcome:

Design the data structure based on the database structures and its characteristics.

Understand the concept of data warehouse, its components and characteristics.

Can Do Statements:

- I can read and write about the history of data warehousing.
- I can write a short paragraph related to the history of data warehousing.
- I can list important information about the history of data warehousing.

Ask your partner:

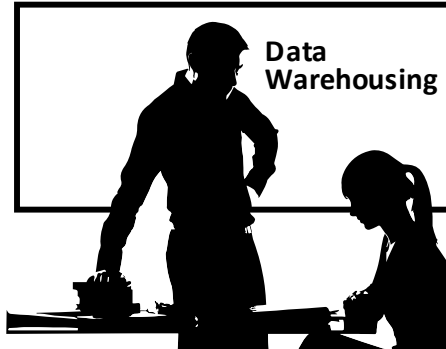
- Is managing information in a business environment easy? Why, or why not?
- What historical facts in the field of computer science do you know?





Let's Explore!

If we understand the history behind a concept, the concept itself becomes clearer



Instructions: Read the text below.

- Natalia:** Sir, I think we need to review some things before the exam begins.
- Instructor:** I agree; you may consider historical information is not important for this job, but it is! If you know the history behind a simple concept like data warehousing, the concept itself becomes clearer and easier to understand.
- Natalia:** Yes! Please, give us a summary of the history of data warehousing, for I don't remember much of it.
- Instructor:** OK, let's go back to the 1980's, databases are not very sophisticated, and engineers are about to discover a big problem: if they optimize the database for transactions, it is not efficient for reporting and data analysis purposes; so, integration with decision support systems becomes very complicated. Two people realize about this problem and come up with a solution: Data warehousing. They are Bill Inmon and Ralph Kimball; the first one coins the term "Data Warehouse" in the 70's. Do you understand the problem they found, and their solution?
- Natalia:** I may be wrong, but let's see: if I have a database that I use for transactions, it's difficult to use the data in it to create reports, unless I use a relational model.
- Instructor:** Exactly. And that relational model is part of the solution popularized in the 1990's with data warehousing applications. In general, the concept of data warehousing refers to saving all data from different sources in a single storage location, until we need to use it. Is that clear?
- Natalia:** Let's see...If we store all information from our data sources in a single tool, can we use it all at once?
- Instructor:** Correct! In 1992, Inmon writes a book called "Building the Data Warehouse", and Kimball publishes "The Data Warehouse Toolkit" in 1996. These two books offer different approaches to relational modeling, yet they are equally good at explaining how to implement data warehousing in business environments.
- Natalia:** I'm sorry, sir; I'm going to write down the titles of the books while you finish your explanation.



Let's Practice!

Let's go over some key information on data warehousing

Instructions: Select the correct option to correctly complete each sentence. Be careful with the distractors.

stores / learn / store / learns / is not / using / uses

1. If we _____ the origins of all processes involved in data warehousing, we are able to understand the importance of our job.
2. The database _____ efficient for reporting and analyzing data if it's optimized for transactions.
3. If we _____ a relational model, it's less difficult to use data in order to create reports.
4. If a business _____ all data in a single storage location, they can use it all at once.



Let's Practice!

Learning the ropes* of data warehousing is essential, so let's read about its origins.

*To "learn the ropes" is to learn how to do something.

Instructions: Use the dates from the box to correctly complete the sentences. Be careful with the distractors:

Nor / for / when the / and / while / But, / but Bill Inmon / when

1. Back in the 1980's databases are not sophisticated ____ efficient for data analysis purposes.
2. Data warehousing is officially born in the 80's, _____ and Ralph Kimball coin the term Data warehouse in the 70'.
3. It was in 1992_____ Inmon wrote "Building the Data Warehouse."
4. _____ it was _____ Ralph Kimball who wrote "The Data Warehouse Toolkit" in 1996.
5. It was during the 90's _____ relational model for data warehousing applications reached the peak.



Extension Activity

Drawing and completing my own timeline about the history of data warehousing

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information on the history of data warehousing.
- Make a list of the most important events and their dates.
- Draw a timeline in which you organize those events.
- Show your timeline to your instructor and prepare a description about the history of data warehousing.

Self-Assessment

Yes

No

- I look for more information about the history of data warehousing.
- I make a list with important events related to data warehousing throughout history.
- I draw a timeline with the information.
- I show my timeline to my instructor and check my pronunciation.

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Target Words

- coin** (verb): to be the first one to use an expression
e.g. Bill Inmon coined the term "Data Warehouse" in the 70's
- summary** (noun): a short repetition of the important aspects of a topic or a text
e.g. Please, give us a summary of the history of data warehousing.
- come up with** (phrasal verb): to suggest or create something
e.g. Two people saw this problem and came up with a solution: Data warehousing.
- source** (noun): the place or location of origin or something
e.g. The concept of data warehousing refers to saving all data from different sources in a single storage location.
- at once** (adverbial expression): this expression has two meanings: "at the same time", and "immediately"
e.g. If we store all information from our data sources in a single tool, can we use it all at once?



Learning Points

Zero Conditional: Review

As you should know already, the general purpose of conditionals is expressing cause-effect or action-consequence relationships.

In the case of the **Zero conditional**, the effect or consequence expressed is considered a truth or a fact. This is the reason we use **Present Simple** in the two clauses that make up this conditional structure:

- **If we *don't implement* a data warehouse in the company, the company *doesn't keep* track of its data.**
- **When you *implement* a data warehouse, you *keep* all your data into one tool.**

The examples above show the basic negative and affirmative structures, respectively. You can also see, from the second example, that the word **when** can replace the word **if** in a conditional sentence. Also, notice the **comma** that separates the two sentences of the conditional. This comma is always necessary when the **if sentence** goes first. But remember, you can also start with the **consequence**, and in this case, **we do not use the comma**:

- *The company **doesn't keep** track of its data **if we don't implement** a data warehouse in the company,*
- *you **keep** all your data into one tool **when you implement** a data warehouse.*

We can also use the zero conditional in **interrogative form** to ask about the consequence of an action when we have an assumption of what the consequence is. Let's compare the following questions:

- *What **happens** if we **update** the database?*
- ***Does** performance **improve** if we **update** the database?*

In the first one, we are not suggesting any possible consequence, and that is evident by the use of what. In the second one, we're using an interrogative phrase in Present Simple (**Does performance improve**) to express our assumption.

Something else to notice: The answer to the first question must be an open-ended one, and the answer to the second will probably be a yes/no answer.

Examples

- *If you know the history behind a simple concept like data warehousing, the concept itself becomes clearer and easier to understand.*
- *If they optimize the database for transactions, it is not efficient for reporting and data analysis purposes.*





Learning Points

Subordinating Conjunctions, Coordinating Conjunctions: Review

We can classify **conjunctions** into two categories, depending on the type of clauses they connect. Knowing this will help you improve your writing.

You probably know this already: a conjunction is a word that connects two ideas in a sentence. Now, depending on the type of relationship that these ideas have, we will connect them using **coordinating** or **subordinating conjunctions**. But, what's the difference? Let's take a look at the following example:

*This software is expensive **but** its features make it the best!*

As you can see, there are two independent parts of this sentence, and they are joined by the conjunction **but**. The difference between these two types of conjunctions is that **coordinating conjunctions** join **two independent clauses**, and **subordinating conjunctions** join **an independent clause with a subordinate one**. We will explain this later, but first, let's take a look at the list of conjunctions, so you can memorize them; look for their definition and share them with your peers:

(CHART 1)

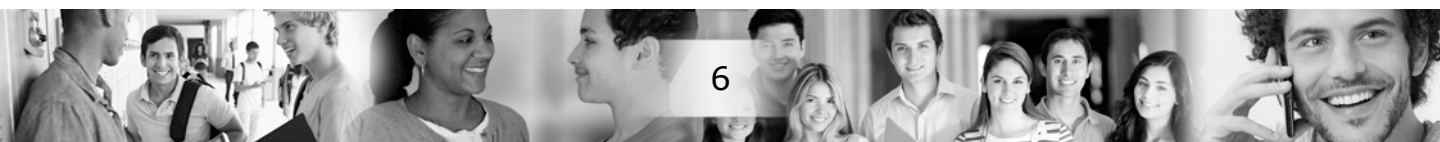
Now that you know the conjunctions, you need to understand what dependent and independent clauses are.

An **independent clause** is a phrase with three characteristics: it has a subject, it contains an action (or predicate), and it expresses a full idea; this means that an independent clause works as a sentence. Let's see some examples:

- *I want to implement a database.*
- *Jane is reading a manual.*
- *I work.*
- *Cleanse the database.*

The examples above are all independent clauses; they don't need additional information to express a complete idea. Now, a **subordinate clause** is a phrase that does not express a complete idea; this may be because there is no subject, or there is no action, or something is missing. Let's see some examples:

- **When** I switch the software.
- **For** she has to implement the new software.
- **While** you are working there.
- **Until** it looks perfect.





Learning Points

These phrases are **dependent** because they do not express a full idea; we are missing some information (What happens when you switch the software? What happens while you are working there?). Let's see the two groups of examples joined together:

- *I want to implement a database **when** I switch the software.*
- *Cleanse the database **until** it looks perfect.*
- *Jane is reading a manual **for** she has to implement the new software.*

Now, both parts of each sentence make sense! The dependent clauses are working as the **complement** of the independent ones. The main idea of the sentence is always located in an independent clause, while the information that the **subordinate clauses** provide is additional, but not necessary for the sentence to make sense.

Now, let's see some examples of two clauses joined by **coordinating conjunctions**:

- *I updated this database **and** all the information was erased!*
- *This computer is old **but** you can open database sheets in it.*
- *I don't like this keyboard, **so** I bought a new one.*

In the examples above, the first and second parts of each sentence express a full idea; there is not emphasis in one of the two parts because the two are equally important for the meaning of the sentence.

To conclude this review: **Use coordinating conjunctions** when you want to express two complete ideas, and **use subordinating conjunctions** when you want to provide additional information to complement the main idea of an independent clause.

CHART 1

Coordinating conjunctions	Subordinating conjunctions
For And Nor But Or Yet So	As When After Because Until Before While Unless ...

Examples

- *Please, give us a summary of the history of data warehousing, for I don't remember much of it.*
- *These two books offer different approaches to relational modeling, yet they are equally good at explaining how to implement data warehousing*
- *Save all data from different sources in a single storage location, until we need to use it.*

